

Article

Effect of Printing Speed and Core Relative Density on the Mass-Specific Bending Performance of 3D-Printed PLA Re-Entrant Sandwich Structures

Dechawat Wannarong^a and Thanyarat Singhanart^{b,*}

Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok 10330, Thailand

E-mail: ^adechawatwannarong@gmail.com, ^{b,*}Thanyarat.S@chula.ac.th (Corresponding author)

Abstract. This study investigates the bending performance of 3D-printed polylactic acid (PLA) re-entrant honeycomb sandwich structures fabricated using fused deposition modelling (FDM). The effects of printing speed and core relative density on mechanical efficiency were examined using a complete 2×3 factorial set of six parameter combinations, arranged in a mixed-level L6 format for signal-to-noise ratio analysis. Three printing speeds (50, 100, and 150 mm/s) and two relative densities (0.3 and 0.5) were considered, while other printing parameters were kept constant. Three-point bending tests were conducted in accordance with ASTM C393, and the mechanical responses were evaluated in terms of maximum load per mass and bending stiffness per mass to account for weight efficiency. Within the investigated range, increasing printing speed enhanced both maximum load per mass and bending stiffness per mass, whereas core relative density had a greater influence on bending stiffness per mass. Signal-to-noise ratio analysis identified printing speed as the dominant factor for maximum load per mass, while relative density was more influential for bending stiffness per mass. These trends were confirmed by analysis of variance and Pareto chart evaluation. Among the six investigated combinations, a printing speed of 150 mm/s and a relative density of 0.3 produced the highest measured mass-specific load capacity and bending stiffness. This combination represents the best-performing condition within the investigated parameter space and should not be interpreted as a global optimum. Additive-model estimates differed from the corresponding measured means by less than 2%; however, this comparison represents within-sample agreement rather than independent experimental validation. Empirical linear models described 86% of the measured response variation; however, these models are intended only to summarize trends within the tested parameter domain. The findings provide range-specific evidence for selecting printing conditions for comparable integrally printed PLA re-entrant sandwich structures under quasi-static bending.

Keywords: PLA, printing speed, 3D printing, sandwich honeycomb, re-entrant, bending strength

ENGINEERING JOURNAL Volume 30 Issue 8

Received 25 December 2025

Accepted 14 August 2026

Published 31 August 2026

Online at <https://engj.org/>

DOI:10.4186/ej.2026.30.8.129

1. Introduction

Honeycomb sandwich panels are widely used in engineering applications such as aerospace, automotive, marine, and civil structures due to their excellent stiffness-to-weight and strength-to-weight ratios. With the advancement of additive manufacturing, it has become possible to fabricate honeycomb structures in which both the core and face sheets are produced from the same polymer material. This fabrication approach eliminates the need for secondary bonding and ensures a fully integrated structure with consistent mechanical behaviour.

Among various honeycomb geometries, re-entrant (auxetic) structures have gained significant attention due to their negative Poisson's ratio, enhanced energy absorption, and improved bending resistance compared to conventional honeycombs. The ability of re-entrant cells to expand laterally under tension and contract under compression makes them highly suitable for lightweight structural applications requiring superior deformation capability.

The emergence of 3D printing, particularly fused filament fabrication (FFF), has enabled the precise manufacturing of re-entrant honeycomb cores with complex geometries that were difficult to achieve using traditional processes. Polylactic acid (PLA) remains one of the most widely used materials in FFF due to its ease of processing, low cost, and environmental friendliness. However, the mechanical performance of 3D-printed PLA structures—including bending strength—depends strongly on the printing parameters selected during fabrication.

Previous studies have focused primarily on the influence of cell geometry, infill density, and material type on the performance of auxetic and honeycomb structures. While some research has examined general printing parameters, the specific effect of printing speed on the bending behaviour of 3D-printed PLA re-entrant honeycomb structures remains insufficiently explored. Printing speed directly affects heat transfer, filament bonding, interlayer adhesion, and overall printing quality; yet, there is limited quantitative data addressing how these factors translate into bending strength, especially when combined with varying core densities.

The limited available evidence motivates a range-bounded investigation of how printing speed and core relative density influence the quasi-static bending response of integrally printed PLA re-entrant sandwich structures. Rather than seeking a global optimum, the present study evaluates the relative importance of the two parameters within a predefined processing and density domain.

The scientific contribution of this study lies in identifying the distinct and interacting roles of printing speed and core relative density in the bending response of an integrally printed PLA re-entrant sandwich structure. In particular, the study distinguishes absolute load capacity from mass-specific mechanical efficiency and shows that an increase in core density does not necessarily improve performance after accounting for the associated mass

increase. The results also quantify how the effect of printing speed differs between the two investigated density levels. The Taguchi S/N analysis, ANOVA, and regression are used as supporting statistical tools rather than being presented as methodological innovations.

2. Literature Review

Additive manufacturing (AM), commonly referred to as 3D printing, has gained significant traction in both research and industrial sectors due to its versatility in fabricating complex geometries that are difficult or impossible to achieve through conventional manufacturing methods. AM enables rapid prototyping and customization with minimal material waste, making it highly attractive for industries such as aerospace, automotive, biomedical, and construction [1]. The process involves the layer-by-layer deposition of materials, typically thermoplastic polymers like Polylactic Acid (PLA) and Acrylonitrile Butadiene Styrene (ABS), to create functional parts with tailored mechanical properties.

Polylactic Acid (PLA) is one of the most widely used polymer materials in fused deposition modelling (FDM) due to its ease of printing, low processing temperature, low toxicity, and bio-based origin [2]. Its biodegradability and dimensional stability make PLA attractive for prototyping, biomedical devices, educational applications, and consumer products. However, despite its advantages, PLA suffers from relatively low thermal resistance, brittleness, and limited impact strength compared with engineering thermoplastics [3]. These limitations reduce its suitability for load-bearing applications or environments requiring high thermal stability. Recent studies highlight the significance of understanding PLA's thermal transitions, crystallinity, and mechanical behaviour to optimize its performance in additive manufacturing [4].

Research on PLA for 3D printing has primarily focused on two major directions: optimizing processing parameters and modifying material formulations. Process parameters can substantially influence the mechanical performance of FDM-printed PLA. In particular, printing speed and extrusion temperature have been shown to affect the tensile properties and dimensional quality of printed PLA parts [5]. Parallel to this, extensive research has explored PLA composites and blends by incorporating fillers such as carbon fibres, natural fibres, nanoparticles, and plasticizers to enhance toughness, thermal stability, and mechanical strength [6]. Recent review papers emphasize that both printing-parameter optimization and material modification are essential in broadening PLA's applications, making it a promising material for sustainable and high-performance additive manufacturing across various engineering fields [4].

The mechanical performance of 3D-printed parts is strongly dependent on several process parameters. Among these, infill density, printing speed, layer thickness, and printing temperature are the most commonly investigated. Studies have shown that increasing infill density leads to

greater stiffness and strength due to higher material occupancy [1,7]. Printing speed influences interlayer bonding—higher speeds may reduce bonding strength and lead to weaker parts [8]. Layer thickness affects surface quality and load-bearing behaviour; thicker layers enhance stiffness but may decrease resolution [1].

In the present study, printing temperature was excluded as a variable. Since Polylactic acid (PLA) exhibits a relatively narrow optimal printing-temperature window, variations outside this range have minimal impact on mechanical performance once adequate interlayer adhesion is achieved [9]. Moreover, recent studies suggest that key geometric and process-related parameters – such as infill density, layer thickness, print speed – exert a larger influence on strength and interlayer bonding quality in FDM-printed PLA than temperature variation alone [10–12]

Sandwich constructions are widely utilized in aerospace, automotive, and marine applications due to their high stiffness-to-weight ratio and energy absorption capacity [13,14]. A typical sandwich panel consists of two strong face sheets bonded to a lightweight core, often a honeycomb or foam structure as shown in Fig.1. The core serves to separate the faces and resist shear loads, while the face sheets primarily withstand tensile and compressive stresses during bending. The mechanical behaviour of sandwich panels, particularly under bending, is influenced by the core geometry, face material, and interfacial bonding quality [14,15].

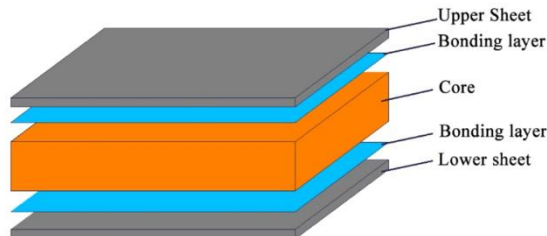


Fig.1. The construction of the sandwich structure.

Several studies have emphasized that an infill density of 100% in the core region provides the highest stiffness and bending resistance, albeit at the expense of weight. For example, Faidallah et al. [16] compared PLA sandwich structures with rhombus and honeycomb cores, reporting superior tensile and bending strengths for the rhombus design. Similarly, Zhang et al. [17] demonstrated that optimized variable-density quasi-honeycomb cores improve bending stiffness, shear resistance, and overall energy absorption under three-point bending tests.

Although density, pattern, material, and temperature have been studied extensively, printing speed has received less systematic attention. Some studies suggest that higher speeds can reduce tensile strength due to inadequate layer bonding [18]. For instance, a 2024 study investigated infill density and printing speed on PLA specimens printed via FDM. Results showed that increasing infill density had a much stronger effect on ultimate tensile strength than speed; increasing speed from 30 to 90 mm/s decreased

ultimate tensile strength by $\sim 5.6\%$, while increasing density from 40% to 80% increased ultimate tensile strength by $\sim 35.2\%$.

Honeycomb core structures have been the focus of extensive research due to their excellent mechanical performance and lightweight characteristics. Studies typically investigate key parameters such as bending stiffness (EI) and maximum load (P) to quantify structural performance under flexural conditions. For example, Montazeri et al. [14] showed that hybrid honeycomb geometries with different Poisson's ratios exhibit enhanced load-carrying capacity and specific energy absorption under bending. Furthermore, the introduction of hierarchical or re-entrant features in the core design can significantly improve both stiffness and deformation behaviour [19,20].

In practical engineering applications, honeycomb sandwich structures are widely used in aerospace panels, automotive components, and lightweight structural systems, where materials such as aluminum honeycombs and Nomex cores are commonly employed. These materials offer excellent stiffness-to-weight ratios but are often associated with higher manufacturing costs and limited design flexibility. Recently, polymer-based honeycomb structures, including PLA, have been explored as cost-effective and lightweight alternatives, particularly for prototyping and non-critical load-bearing applications. Compared with traditional honeycomb cores, 3D-printed structures provide greater geometric flexibility and enable the fabrication of complex architectures such as re-entrant (auxetic) designs. However, their bending performance and mechanical efficiency, especially under varying printing parameters, remain insufficiently understood. These limitations highlight the need for systematic studies focusing on the bending efficiency of 3D-printed honeycomb sandwich structures under controlled printing parameters.

Most existing studies on 3D-printed sandwich structures have primarily examined the influence of core geometry and infill density on mechanical behaviour, with predominant focus on tensile properties evaluated using standard dogbone-shaped specimens. In contrast, investigations into bending performance—particularly for sandwich structures incorporating honeycomb cores—remain relatively limited. Furthermore, the effect of printing speed, a key process parameter in additive manufacturing, on the flexural response of such structures has not yet been systematically explored. To address these research gaps, the present study investigates the bending behaviour of 3D-printed PLA sandwich structures featuring re-entrant honeycomb cores. The specimens were fabricated using fused deposition modelling (FDM) at three printing speeds and two relative-density levels, enabling a controlled comparison of the effects of these parameters on flexural performance within the selected experimental range. The findings provide range-specific insight into the effects of printing speed and core relative density on the bending response of the investigated re-entrant sandwich structures. Accordingly, the study is

positioned as a screening investigation of parameter effects rather than a comprehensive optimization of re-entrant geometry or FDM processing conditions.

3. Materials and Methods

The focus is on the geometric characteristics and mechanical considerations of the re-entrant honeycomb structure, selected for its distinctive auxetic behaviour and enhanced energy absorption capability. The re-entrant configuration is characterized by inwardly pointed cell walls, resulting in a negative Poisson's ratio that enables unique deformation patterns under load. The schematic diagram illustrating the microstructure of the re-entrant lattice is depicted in Fig. 2. The relative density of re-entrant honeycomb configuration, a key parameter influencing the mechanical performance and weight efficiency of the structure, can be calculated using specific geometric relationships and formulas tailored to each design. The re-entrant honeycomb, known for its auxetic properties (negative Poisson's ratio) taking into account the angles and dimensions of its inwardly pointed walls. The relative density of the re-entrant honeycomb structure can be calculated by using Eq. (1).

$$\frac{\rho^*}{\rho_s} = \frac{(t/L)[(H/L) + 2]}{2\cos\theta[(H/L) + \sin\theta]} \quad (1)$$

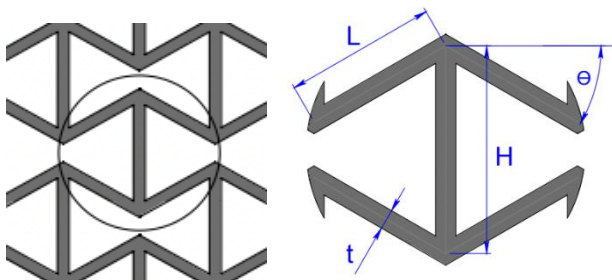


Fig. 2. The schematic of the re-entrant lattice structure.

In this study, a relative density of 0.3 and 0.5 are considered with the unit cell dimension of 16 mm x 16 mm. These levels were selected to represent two separated lightweight-core configurations while maintaining printability and geometric integrity. Because only two levels were examined, the experiment can estimate the average contrast between these densities but cannot establish nonlinearity, an intermediate optimum, or a global density trend. Table 1 provides the detailed parameters outlining the specific dimensions and characteristics that achieve the desired relative density.

Table 1. Design parameter of re-entrant honeycomb.

$\frac{\rho^*}{\rho_s}$	Re-entrant Honeycomb			
	L (mm)	H (mm)	t (mm)	θ
0.3	9.24	12.62	1.23	30°
0.5	9.24	12.62	2.06	30°

The bending test specimens are prepared in accordance with ASTM C393 standard. This study focuses on evaluating the mechanical properties of specimens produced at three distinct printing speeds: 50 mm/s, 100 mm/s and 150 mm/s. To ensure consistency and comparability, the temperature, infill percentage, and infill pattern were kept constant across all specimens. A 100% infill with a rectilinear pattern was used, and the raster orientation was aligned along the longitudinal direction of the specimen. This configuration ensures that the filament deposition direction is parallel to the primary loading direction during bending, thereby minimizing anisotropic effects and improving load transfer consistency. The sandwich beams used in the bending tests have precise dimensions of 200 mm in length, 75 mm in width, and 12 mm in thickness. Figure 3 provides a schematic depiction of the sandwich beam, illustrating the overall structure and specific details. The bending test specimens had dimensions of 200 mm x 75 mm, with a face sheet thickness of 1 mm, a core height of 10 mm, and a total thickness of 12 mm.

Both the face sheets and the honeycombs core of the sandwich beams are fabricated using Polylactic Acid (PLA) materials, employing a 3D printer to achieve accurate and repeatable production. Both the face sheets and the core are printed simultaneously as a single piece. This approach minimizes the effects of bonding between layers, which can otherwise act as weak interfaces and reduce overall mechanical performance. Printing the face sheets and core together ensures a more integrated structure, enhancing the load transfer and structural integrity of the sandwich beams.

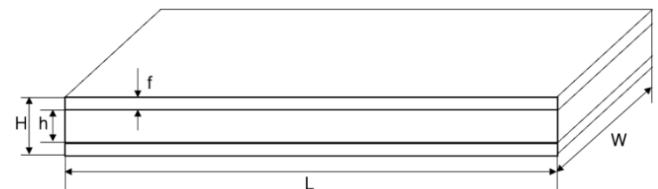


Fig. 3. The schematic of the sandwich beam.

The experimental programme was designed to examine two control factors: core relative density and printing speed. Core relative density was investigated at two levels, 0.3 and 0.5, whereas printing speed was investigated at three levels, 50, 100, and 150 mm/s. These factor levels produced a complete 2x3 factorial set comprising six experimental conditions. The six combinations were organized in a mixed-level L6 format to facilitate signal-to-noise (S/N) ratio analysis. Because all possible combinations of the selected factor levels were tested, the L6 arrangement did not reduce the number of experimental conditions relative to the corresponding full-factorial design. The purpose of the experimental design was therefore to compare the effects of the selected factor levels within the investigated parameter space, rather than to establish a global optimum. The selected factors and levels are presented in Table 3, and the six experimental conditions are summarized in Table 4.

All specimens were manufactured using a Bambu Lab P1S 3D printer. Figure 4 illustrates the fabrication of the re-entrant honeycomb structure during the 3D printing process. Following fabrication, the mass of each specimen was recorded. For each experimental condition, three replicate samples were produced and subsequently subjected to flexural testing. The measured responses were averaged, and standard deviations were calculated to ensure statistical reliability and reduce uncertainties in the reported data.

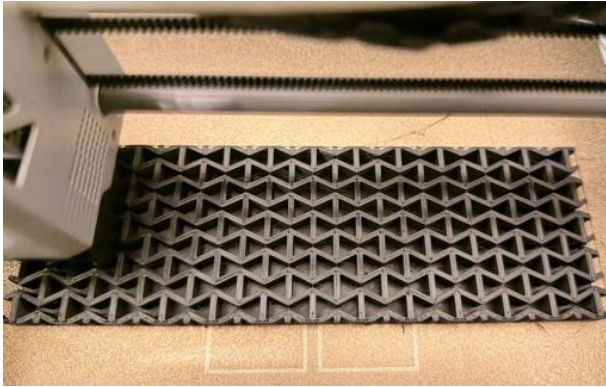


Fig. 4. Top view of the core during the 3D printing process.

Flexural testing was conducted in accordance with the American Society for Testing and Materials (ASTM) C393 standard, which provides a systematic procedure for evaluating the bending behaviour of sandwich structures. Each specimen was positioned centrally between an upper loading indenter and two lower cylindrical supports, forming a standardized three-point bending configuration. After securing the specimen, the universal testing machine was calibrated to establish a precise zero reference point. A vertical load was then applied at the mid-span of the specimen at an initial rate of 6 mm/min, with a support span of 150 mm. This configuration enables accurate determination of the flexural stiffness and strength of the sandwich constructions.

The experimental responses were evaluated using maximum load per mass and bending stiffness per mass to account for differences in specimen mass. For each experimental condition, the mean response and standard deviation were calculated from three replicate specimens. The Taguchi “larger-the-better” criterion was used to calculate the signal-to-noise (S/N) ratio according to Equation (2), where n is the number of replicate specimens and y_i is the measured response of the i th specimen. A higher S/N ratio indicates a combination of a larger response and lower variability among replicate measurements, following the statistical procedures commonly employed in parameter-screening and optimization studies [21,22].

Analysis of variance (ANOVA) was subsequently performed to quantify the contributions of printing speed, core relative density, and their interaction to the variation in each response. Statistical significance was evaluated at a

95% confidence level, corresponding to $\alpha = 0.05$, and an effect was considered statistically significant when $p < 0.05$. The percentage contribution of each factor was calculated from its sequential sum of squares relative to the total sum of squares. Pareto charts of the standardized effects were also constructed to provide a graphical comparison of the relative magnitudes of the main and interaction effects. The combined use of experimental-design and statistical-modelling methods is consistent with established parameter-analysis frameworks [21–23].

Empirical linear regression models were additionally fitted to summarize the relationships between the investigated factors and the measured responses. Because core relative density was examined at only two levels, its regression coefficient represents the average contrast between relative densities of 0.3 and 0.5 rather than confirmation of a continuous linear density relationship. Similarly, because 50 and 150 mm/s represent the boundaries of the investigated printing-speed range, the regression models were not used to extrapolate responses outside the range of 50–150 mm/s. The models should therefore be interpreted as empirical descriptions of the measured trends within the investigated parameter domain, rather than as generally applicable predictive equations.

It should be emphasized that the terms “best-performing condition” and “selected condition” used in this study refer only to the six investigated parameter combinations. The experimental and statistical analyses do not establish a global optimum because only two core relative-density levels and three printing-speed levels were examined.

Therefore, the comparison of core relative density is limited to the two investigated levels of 0.3 and 0.5. The results identify the better-performing level between these two conditions but do not establish a continuous density trend or a global optimum. Additional intermediate and extended density levels are required for a more comprehensive evaluation of the influence of core relative density.

Table 3. Printing parameters and their corresponding levels.

Parameter	Level 1	Level 2	Level 3
Relative density	0.3	0.5	-
Printing speed (mm/s)	50	100	150

$$\frac{S}{N_{max}} = -10 \log_{10} \left(\frac{1}{n} \sum_{i=1}^n \frac{1}{y_i^2} \right) \quad (2)$$

Table 4. Complete 2×3 factorial combinations arranged in a mixed-level L6 format.

No	Relative Density	Printing speed (mm/s)
1	0.3	50
2	0.3	100
3	0.3	150
4	0.5	50
5	0.5	100
6	0.5	150

4. Results and Discussion

This section presents the experimental results and statistical analyses of the bending performance of 3D-printed PLA re-entrant honeycomb sandwich structures. The effects of printing speed and relative density are evaluated based on both strength- and stiffness-related responses obtained from three-point bending tests. To enable a meaningful comparison between specimens with different masses, the mechanical responses are normalized by specimen mass, and the Taguchi signal-to-noise (S/N) ratio using the larger-the-better criterion is employed to assess the influence of process parameters and identify the best-performing factor levels within the investigated parameter space.

Table 5 summarizes the mean maximum load, specimen mass, maximum load per mass, standard deviation, and corresponding signal-to-noise (S/N) ratio for each experimental condition. Maximum load represents the absolute load-carrying capacity of the sandwich beam, whereas maximum load per mass represents its structural efficiency after accounting for differences in specimen mass. These two responses should be interpreted separately because a configuration with a higher absolute load capacity does not necessarily provide better lightweight efficiency.

At both investigated relative-density levels, maximum load per mass increased with printing speed. At a relative density of 0.3, increasing the printing speed from 50 to 150 mm/s increased the maximum load per mass from 9.02 to 11.46 N/g, corresponding to an improvement of 27.1%. At a relative density of 0.5, the corresponding response increased from 8.76 to 10.10 N/g, representing an improvement of 15.3%. The different magnitudes of improvement at the two density levels indicate that the effect of printing speed depended partly on core relative density. This observation is consistent with the speed–density interaction subsequently identified by ANOVA.

The beneficial effect of printing speed observed in the present study differs from the reductions in tensile strength reported in some previous investigations, where higher printing speeds were associated with insufficient interlayer bonding [8,18]. This difference may arise because the present specimens were integrally printed sandwich structures subjected to bending rather than homogeneous tensile specimens. Within the investigated speed range, the higher printing speeds may have promoted more continuous filament deposition and

effective load transfer along the deposited paths without causing sufficient bonding deterioration to reduce the measured bending responses. Nevertheless, the present results do not demonstrate that increasing printing speed beyond 150 mm/s would continue to improve performance. Excessively high printing speeds may increase porosity, dimensional deviations, and insufficient interlayer bonding; therefore, the observed trend should be restricted to the investigated range of 50–150 mm/s.

Core relative density had different effects on absolute load capacity and mass-specific performance. At 50 mm/s, increasing relative density from 0.3 to 0.5 increased the mean maximum load from 779.17 to 1038.54 N, corresponding to an increase of 33.3%. At 100 mm/s, the maximum load increased from 927.09 to 1108.34 N, representing an increase of 19.6%. At 150 mm/s, it increased from 992.71 to 1194.80 N, corresponding to an increase of 20.4%. These results demonstrate that the higher-density core provided greater absolute load-carrying capacity at all three printing speeds.

However, increasing relative density also produced a substantial increase in specimen mass. At printing speeds of 50, 100, and 150 mm/s, the mean mass increased by approximately 37.2%, 35.7%, and 36.6%, respectively, when relative density increased from 0.3 to 0.5. Because these mass increases were greater than the corresponding gains in maximum load, the maximum load per mass decreased. The reductions in maximum load per mass were approximately 2.9% at 50 mm/s, 12.0% at 100 mm/s, and 11.9% at 150 mm/s. Thus, the higher-density specimens were stronger in terms of absolute maximum load, whereas the lower-density specimens were generally more efficient on a mass-normalized basis.

Figure 5 presents the bending stiffness per mass, and Figure 6 shows the corresponding S/N ratios for the investigated printing speeds and relative densities. Bending stiffness per mass increased with printing speed at both density levels. The highest measured bending stiffness per mass, 2.08 N/mm/g, was obtained at a relative density of 0.3 and a printing speed of 150 mm/s. The S/N-ratio trends shown in Figure 6 were consistent with the measured stiffness responses, with the highest S/N ratio occurring at 150 mm/s. Among the two density levels, the lower-density specimens provided higher mass-specific stiffness because the reduction in specimen mass compensated for their lower material content. This result should not be interpreted as evidence that a relative density of 0.3 maximizes stiffness over a continuous density range; it identifies only the better-performing level between the two densities investigated.

Overall, the results demonstrate a trade-off between absolute load capacity and lightweight efficiency. Increasing core relative density increased the absolute maximum load, whereas the lower relative density generally provided higher maximum load per mass and bending stiffness per mass. Printing speed improved both mass-specific responses within the investigated range, although the magnitude of improvement depended on relative density. These findings indicate that parameter

selection should be based on the intended design requirement: a higher-density core may be preferred when absolute load capacity is the primary objective, whereas a lower-density core may be advantageous when structural mass is a critical constraint. These conclusions are restricted to the investigated relative-density levels, printing-speed range, specimen geometry, and quasi-static bending conditions.

During the three-point bending tests, the specimens exhibited global bending accompanied by localized deformation beneath the loading nose. Macroscopic observations indicated tensile-side damage together with progressive deformation of the re-entrant core. Local bending, rotation, and buckling of the inclined cell walls were also observed near the central loading region. These deformation modes are consistent with the load-transfer mechanism of sandwich beams, in which the face sheets primarily resist bending stresses while the core transfers shear and maintains separation between the face sheets [14,19,20].

The higher-density cores carried greater absolute maximum loads, which may be associated with the greater bending resistance of their thicker cell walls. In contrast, the lower-density cores provided higher mass-specific performance despite their thinner and more deformable cell walls. Printing speed may additionally affect the response through filament continuity and interlayer bonding; however, the present measurements do not isolate these microscale effects.

Because post-failure imaging, in-situ deformation measurement, and microscopic fracture examination were not included, the precise sequence of damage initiation could not be conclusively determined. Therefore, the failure mechanisms discussed here should be regarded as interpretations based on macroscopic observations and established sandwich-beam mechanics rather than as direct microscopic evidence.

To place the present results in a quantitative context, increasing printing speed from 50 to 150 mm/s improved maximum load per mass by 27.1% at a core relative density of 0.3 and by 15.3% at a core relative density of 0.5. In comparison, Faidallah et al. [16] reported that a rhombus-core PLA sandwich exhibited 39.8% higher bending strength than their honeycomb-core configuration. Montazeri et al. [14] reported specific-energy-absorption improvements of 136.9% and 475.1% for selected zero-Poisson-ratio hybrid honeycombs relative to their benchmark configurations. These comparisons demonstrate that both processing parameters and core geometry can produce substantial changes in bending and lightweight performance. However, direct comparison of absolute maximum loads, stiffness values, or mass-specific responses is inappropriate because the studies differ in material, specimen dimensions, core geometry, face-sheet configuration, support span, and performance definitions. Accordingly, the comparison is based on within-study percentage changes and is not intended to establish direct performance superiority of the present specimens.

Table 5. Maximum load values with the calculated S/N ratios.

No	Max. load (N)	Mass (g)	Max. load per mass (N/g)	SD	S/N ratio
1	779.17	86.43	9.02	0.04	19.10
2	927.09	86.80	10.68	0.15	20.57
3	992.71	86.62	11.46	0.12	21.18
4	1038.54	118.59	8.76	0.31	18.85
5	1108.34	117.80	9.40	0.38	19.46
6	1194.80	118.28	10.10	0.31	20.09

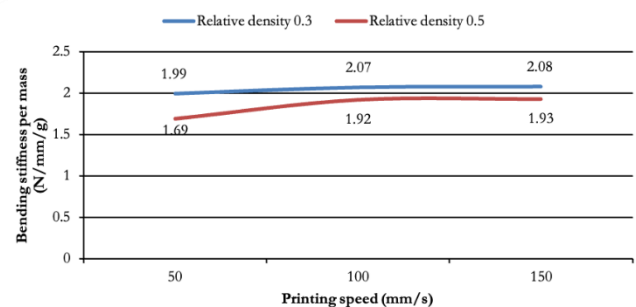


Fig. 5. Effect of printing speed and relative density on bending stiffness per mass.

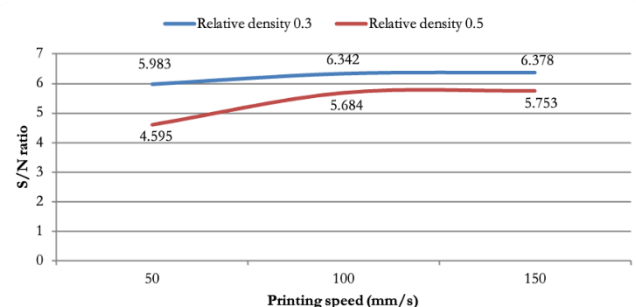


Fig. 6. Effect of printing speed and relative density on S/N ratio of bending stiffness per mass.

Table 6 presents the response table for the S/N ratios of maximum load per mass. For printing speed, the mean S/N ratio increased from 18.97 at 50 mm/s to 20.64 at 150 mm/s. The delta value of 1.66 was larger than that of core relative density, indicating that printing speed had the greater influence on maximum load per mass within the investigated factor levels. For core relative density, the mean S/N ratio decreased from 20.29 at a relative density of 0.3 to 19.46 at 0.5, with a delta value of 0.82. Thus, 150 mm/s and a relative density of 0.3 were associated with the highest S/N ratios among the investigated levels. This combination represents the best-performing factor levels within the tested parameter space and should not be interpreted as a global optimum. Figure 7 presents the percentage contributions of the investigated parameters to maximum load per mass. Printing speed had the largest contribution, consistent with the S/N-response analysis in Table 6 and the ANOVA results in Table 8.

The dominance of printing speed observed in the S/N response is consistent with the role of process parameters in fused deposition modelling, where deposition dynamics and interlayer continuity strongly influence load-bearing capability.

Table 7 presents the response table for the S/N ratios of bending stiffness per mass. Core relative density exhibited a slightly larger delta value than printing speed, with values of 0.89 and 0.78, respectively. This indicates that bending stiffness per mass was more sensitive to the difference between the two investigated relative-density levels than to the investigated printing-speed levels. The highest S/N ratios were obtained at a relative density of 0.3 and a printing speed of 150 mm/s. This combination was therefore identified as the best-performing investigated condition for mass-specific bending stiffness. Because only two relative-density levels were examined and 150 mm/s was the upper boundary of the printing-speed range, these results do not establish a continuous density trend or a global optimum. Figure 8 presents the percentage contributions of the investigated parameters to bending stiffness per mass. Core relative density had the largest contribution, consistent with the S/N-response analysis in Table 7 and the ANOVA results in Table 9.

The difference in factor ranking between Tables 6 and 7 indicates that maximum load per mass and bending stiffness per mass were influenced differently by the investigated parameters. Printing speed had the greater influence on maximum load efficiency, whereas core relative density had the greater influence on mass-specific bending stiffness. This distinction suggests that strength- and stiffness-related responses should be considered separately when selecting printing and structural parameters.

To quantitatively evaluate the effects of the investigated parameters, ANOVA was performed for both mass-specific performance responses. The ANOVA results for maximum load per mass are presented in Table 8. Printing speed accounted for 64.21% of the total variation and exhibited the highest F-value, identifying it as the dominant factor within the investigated parameter range. Core relative density contributed 24.70%, while the interaction between printing speed and relative density contributed 6.67%. Printing speed, relative density, and their interaction were statistically significant at the 5% level ($p < 0.05$). These results support the experimental observation that the magnitude of the printing-speed effect differed between the two relative-density levels.

For bending stiffness per mass, the ANOVA results in Table 9 show that core relative density had the largest contribution, accounting for 53.06% of the total variation. Printing speed was the second most influential factor, with a contribution of 31.91%. The speed–density interaction accounted for 6.52% of the variation. Although the interaction contribution was comparatively small, its p-value of 0.033 was below the significance level of 0.05. The interaction was therefore statistically significant but secondary in magnitude relative to the main effects of relative density and printing speed.

Table 6. Response table for the S/N ratios of the maximum load per mass values (S/N: Larger is better).

Level	Printing speed (mm/s)	Relative density
1	18.97	20.29*
2	20.02	19.46
3	20.64*	-
Delta	1.66	0.82
Rank	1	2

*Level associated with the highest S/N ratio

Table 7. Response table for the S/N ratios of the bending stiffness per mass values (S/N: Larger is better).

Level	Printing speed (mm/s)	Relative density
1	5.29	6.23*
2	6.01	5.34
3	6.07*	-
Delta	0.78	0.89
Rank	2	1

*Level associated with the highest S/N ratio

To further illustrate the relative magnitudes of the investigated effects, Pareto charts of the standardized effects were constructed for maximum load per mass and bending stiffness per mass, as shown in Figures 9 and 10, respectively. The charts were interpreted together with the ANOVA results in Tables 8 and 9, which provide the corresponding p-values and percentage contributions.

Figure 9 presents the Pareto chart for maximum load per mass. Printing speed (Factor A) had the largest standardized effect and exceeded the reference line corresponding to the 95% confidence level. This result is consistent with Table 8, in which printing speed accounted for 64.21% of the total variation. Core relative density (Factor B) had the second-largest effect and was also statistically significant. The speed–density interaction (A×B) had a smaller but statistically significant effect, with a contribution of 6.67%. This interaction is consistent with the different improvements in maximum load per mass observed at the two density levels: increasing printing speed from 50 to 150 mm/s produced improvements of 27.1% and 15.3% at relative densities of 0.3 and 0.5, respectively.

Table 8. Analysis of variance (ANOVA) results for maximum load per mass values.

Factor	DoF	Seq SS	Contribution (%)	MS	F-value	P-value
Speed (A)	2	10.939	64.210	5.469	87.309	0.00000007
Relative density (B)	1	4.208	24.704	4.208	67.183	0.00000293
A x B	2	1.136	6.673	0.568	9.073	0.00397728
Error	12	0.751	4.412	0.062	--	--
Total	17	17.036	100	--	--	--

Table 9. Analysis of variance (ANOVA) results for bending stiffness per mass values.

Factor	DoF	Seq SS	Contribution (%)	MS	F-value	P-value
Speed (A)	2	0.104	31.911	0.052	22.486	0.000087
Relative density (B)	1	0.173	53.056	0.173	74.773	0.000001
A x B	2	0.021	6.516	0.010	4.592	0.033037
Error	12	0.027	8.514	0.002	--	--
Total	17	0.327	100	--	--	--

Figure 10 presents the Pareto chart for bending stiffness per mass. Core relative density (Factor B) had the largest standardized effect, consistent with its contribution of 53.06% in Table 9. Printing speed (Factor A) had the second-largest effect, accounting for 31.91% of the total variation. The speed–density interaction (A×B) had a substantially smaller contribution of 6.52%. Nevertheless, ANOVA indicated that the interaction was statistically significant at the 5% level ($p=0.033$). The interaction should therefore be interpreted as statistically detectable but secondary in magnitude compared with the main effects of relative density and printing speed.

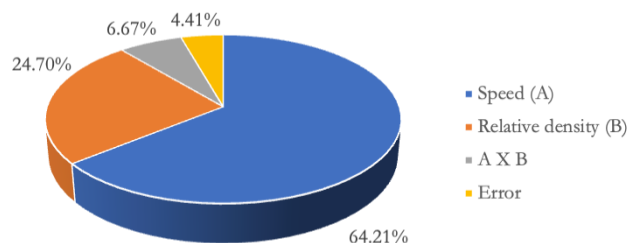


Fig 7. Effect of parameters on maximum load per mass as a percentage.

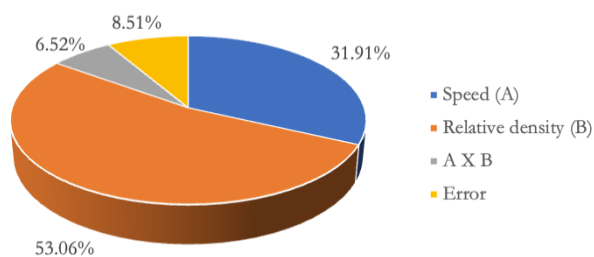


Fig 8. Effect of parameters on bending stiffness per mass as a percentage.

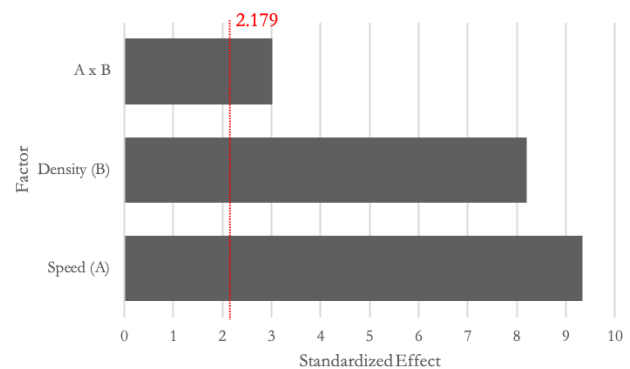


Fig 9. Pareto chart showing the standardized effects of the investigated parameters on maximum load per mass.

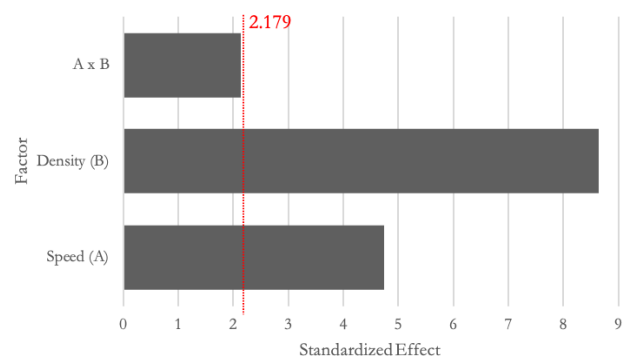


Fig 10. Pareto chart showing the standardized effects of the investigated parameters on bending stiffness per mass.

Based on the measured responses and S/N-ratio analysis, a printing speed of 150 mm/s and a core relative density of 0.3 were identified as the best-performing factor levels among the six investigated combinations. The responses at this condition were estimated using the Taguchi additive model and compared with the corresponding measured means, as summarized in Table 10.

The additive-model estimate of maximum load per mass was 11.28 N/g, compared with the measured mean of 11.46 N/g, corresponding to a difference of 1.57%. For bending stiffness per mass, the model estimate was 2.11 N/mm/g, compared with the measured mean of 2.08 N/mm/g, corresponding to a difference of 1.44% based on the displayed values. This agreement indicates that the additive model reproduced the measured responses at the selected tested condition. However, because the same experimental dataset was used to identify the factor effects and evaluate the model agreement, this comparison represents within-sample agreement rather than independent experimental validation.

The selected combination should not be interpreted as a global optimum. Only two relative-density levels were investigated, and 150 mm/s was the upper boundary of the printing-speed range. Additional density levels and a wider printing-speed range would be required to determine whether another combination provides better performance.

Table 10. Comparison between additive-model estimates and measured responses at the best-performing investigated condition.

Response	Additive model estimate	Measured Mean	Difference (%)
Maximum load per mass (N/g)	11.28	11.46	1.57
Bending stiffness per mass (N/mm/g)	2.11	2.08	1.44

Selected condition: printing speed = 150 mm/s and core relative density = 0.3

To further summarize the relationships between the investigated parameters and the mass-specific bending responses, empirical linear regression models were fitted for maximum load per mass and bending stiffness per mass, as presented in Table 11. Unlike the S/N-response analysis, which compares discrete factor levels, the regression models describe the average response trends within the investigated parameter domain. Because core relative density was examined at only two levels, its regression coefficient represents the average contrast between 0.3 and 0.5 rather than confirmation of a

continuous linear density relationship. The models were therefore not used for extrapolation beyond the tested factor levels.

For maximum load per mass, the regression model produced a positive coefficient for printing speed and a negative coefficient for core relative density. Within the investigated domain, this result is consistent with the observed increase in maximum load per mass with printing speed and the lower mass-specific response of the higher-density specimens. The coefficient of determination, $R^2 = 86\%$, indicates that the model described 86% of the variation in the measured data. However, this goodness-of-fit value represents within-sample descriptive agreement and should not be interpreted as independent predictive validation.

The regression model for bending stiffness per mass also produced a positive coefficient for printing speed and a negative coefficient for core relative density. These coefficient signs are consistent with the measured trends within the investigated conditions. The corresponding value of $R^2=86\%$ indicates similar within-sample descriptive agreement. Nevertheless, the model cannot establish whether bending stiffness varies linearly with relative density because only two density levels were tested.

Figures 11 and 12 compare the fitted values from the empirical linear models with the corresponding measured responses. The fitted values generally follow the experimental trends, with deviations arising from experimental scatter, material heterogeneity, manufacturing variability, and effects not represented by the first-order models. Because the same observations were used to fit and assess the models, the figures demonstrate goodness of fit within the measured dataset rather than out-of-sample predictive accuracy. Independent test conditions and additional density levels would be required to evaluate the predictive performance of the models.

Overall, printing speed and core relative density influenced absolute and mass-specific bending performance in different ways. Increasing printing speed improved both maximum load per mass and bending stiffness per mass within the tested range, whereas increasing core relative density improved absolute maximum load but reduced mass-specific efficiency. Among the six investigated combinations, 150 mm/s and a relative density of 0.3 produced the highest measured mass-specific responses. This condition represents the best-performing tested combination and is not a recommendation beyond the investigated specimen configuration and parameter range.

Table 11. Empirical linear regression equations for the mass-specific bending responses within the investigated parameter domain.

Maximum load per mass = $9.94190 + (0.01894 * \text{printing speed}) - (4.83632 * \text{relative density})$ R-sq=86%
Bending stiffness per mass = $2.17795 + (0.00167 * \text{printing speed}) - (0.98326 * \text{relative density})$ R-sq=86%

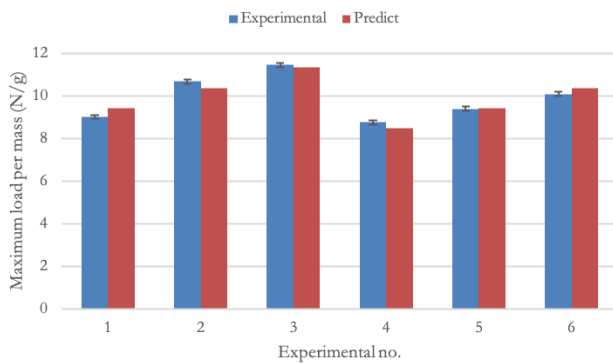


Fig 11. Fitted versus measured maximum load per mass. Error bars represent the standard deviation of the experimental measurements.

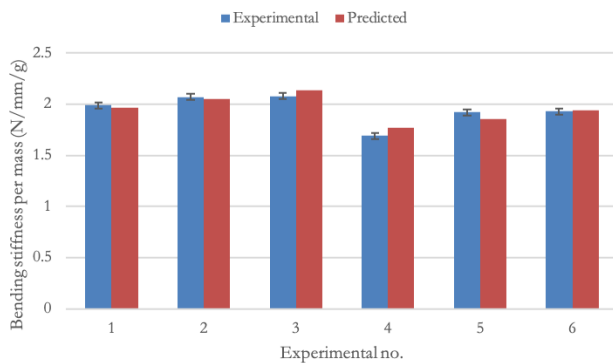


Fig 12. Fitted versus measured bending stiffness per mass. Error bars represent the standard deviation of the experimental measurements.

5. Scope and Limitations

This study was conducted as a screening investigation within a restricted parameter space. Only two core relative-density levels, three printing speeds, one re-entrant cell geometry and size, one raster orientation, one PLA feedstock, one printer configuration, and one sandwich-beam geometry were examined. The mechanical evaluation was limited to quasi-static three-point bending. Consequently, the results cannot establish nonlinear density effects, an intermediate or global optimum, scale effects, or performance under other loading and environmental conditions. The regression equations are empirical summaries of the tested conditions and should not be extrapolated. Furthermore, the failure observations were limited to macroscopic deformation noted during testing because post-failure imaging, in-situ deformation measurement, and microscopic fracture characterization were not available. The conclusions are therefore restricted to the investigated specimen configuration and parameter ranges.

6. Conclusion

This study examined the effects of printing speed and core relative density on the quasi-static bending response of integrally printed PLA re-entrant sandwich beams.

Within the investigated conditions, increasing printing speed from 50 to 150 mm/s increased maximum load per mass by 27.1% at a relative density of 0.3 and by 15.3% at a relative density of 0.5. Printing speed accounted for the largest proportion of variation in maximum load per mass, whereas core relative density had the largest contribution to bending stiffness per mass.

Increasing core relative density from 0.3 to 0.5 increased absolute maximum load by approximately 20–33%, depending on printing speed. However, specimen mass increased by approximately 36–37%, resulting in lower mass-specific performance at the higher density. Absolute load capacity and lightweight efficiency should therefore be evaluated separately when selecting a core configuration.

Among the six investigated combinations, a printing speed of 150 mm/s and a core relative density of 0.3 produced the highest measured maximum load per mass and bending stiffness per mass. This condition represents the best-performing combination only within the tested parameter domain. Because only two density levels were examined and 150 mm/s was the upper boundary of the printing-speed range, the results do not establish a global density optimum or an optimum printing speed beyond the investigated range.

The findings provide range-specific insight into the process–structure–performance relationship of the investigated PLA re-entrant sandwich beams. Further studies should include additional density levels, a wider printing-speed range, other specimen scales and loading modes, and quantitative in-situ or microscopic failure characterization before the findings are generalized to broader structural applications.

Acknowledgement

The presents study was financially supported by Thailand and research funding provided through the Composite Structures Research Unit, Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Thailand.

References

- [1] M. Benamira, N. Benhassine, A. Ayad, and A. Dekhane, “Investigation of printing parameters effects on mechanical and failure properties of 3D printed PLA,” *Eng. Fail. Anal.*, vol. 148, Art. no. 107218, 2023.
- [2] S. Farah, D. G. Anderson, and R. Langer, “Physical and mechanical properties of PLA, and their functions in widespread applications—A comprehensive review,” *Advanced Drug Delivery Reviews*, vol. 107, pp. 367–392, 2016.
- [3] J. A. Travieso-Rodríguez *et al.*, “Mechanical Properties of 3D-Printing Polylactic Acid Parts subjected to Bending Stress and Fatigue Testing,” *Materials*, vol. 12, no. 23, Art. no. 3859, 2019.

- [4] L. Sandanamsamy *et al.*, “A comprehensive review on fused deposition modelling of polylactic acid,” *Polymers*, vol. 8, pp. 775–799, 2022.
- [5] A. A. Ansari and M. Kamil, “Effect of print speed and extrusion temperature on properties of 3D printed PLA using fused deposition modeling process,” *Mater. Today Proc.*, vol. 45, pp. 5462–5468, 2021.
- [6] L. P. Muthe, K. L. Pickering, and C. Gauss, “A Review of 3D/4D Printing of Poly-Lactic Acid Composites with Bio-Derived Reinforcements,” *Composites Part C: Open Access*, vol. 8, Art. no. 100271, 2022.
- [7] V. Acanfora, A. Sellitto, A. Russo, M. Zarrelli, and A. Riccio, “Experimental investigation on 3D printed lightweight sandwich structures for energy absorption aerospace applications,” *Aersp. Sci. Technol.*, vol. 137, Art. no. 108276, Jun. 2023.
- [8] M. R. Khosravani and T. Reinicke, “Effects of raster layup and printing speed on strength of 3D-printed structural components,” *Procedia Struct. Integr.*, vol. 28, pp. 720–725, Jan. 2020.
- [9] M.-H. Hsueh, C.-J. Lai, S.-H. Wang, Y.-S. Zeng, C.-H. Hsieh, C.-Y. Pan, and W.-C. Huang, “Effect of printing parameters on the thermal and mechanical properties of 3D-printed PLA and PETG, using fused deposition modeling,” *Polymers*, vol. 13, no. 11, Art. no. 1758, May 2021.
- [10] S. Ekşi and C. Karakaya, “Effects of process parameters on tensile properties of 3D-printed PLA parts fabricated with the FDM method,” *Polymers*, vol. 17, no. 14, Art. no. 1934, 2025.
- [11] T. Gajjar, R. Yang, L. Ye, and Y. X. Zhang, “Effects of key process parameters on tensile properties and interlayer bonding behavior of 3D printed PLA using fused filament fabrication,” *Prog. Addit. Manuf.*, vol. 10, pp. 1261–1280, Jul. 2024.
- [12] M. N. Ahmad, M. R. Ishak, and M. Z. Zulkafle Hannah, “The Effect of FDM Parameters on the Mechanical Properties of PLA Printed Parts,” *J. Adv. Res. Appl. Mech.*, vol. 123, no. 1, pp. 238–246, Nov. 2024.
- [13] S. Xie, Z. Feng, H. Zhou, and D. Wang, “Three-point bending behavior of Nomex honeycomb sandwich panels: Experiment and simulation,” *Mech. Adv. Mater. Structures*, vol. 28, no. 18, pp. 1917–1931, Jan. 2020.
- [14] A. Montazeri, A. Hasani, and M. Safarabadi, “Bending performance and failure mechanism of 3D-printed hybrid geometry honeycombs with various Poisson’s ratios,” *J. Sandw. Struct. Mater.*, vol. 25, no. 7, pp. 709–729, Aug. 2023.
- [15] M. Sun, D. Wowk, C. Mechefske, and I. Y. Kim, “An analytical study of the plasticity of sandwich honeycomb panels subjected to low-velocity impact,” *Compos. Part B: Eng.*, vol. 168, pp. 121–128, Jul. 2019.
- [16] R. F. Faidallah, M. M. Hanon, Z. Szakál, and I. Oldal, “Study of the mechanical characteristics of sandwich structures FDM 3D-printed,” *Acta Polytechnica Hungarica*, vol. 20, no. 6, pp. 7–26, Nov. 2023.
- [17] X. Zhang, C. Xu, W. Li, and Z. Su, “Study on the bending and shear properties of quasi-honeycomb sandwich structures considering the variable-density core design,” *Compos. Struct.*, vol. 324, Art. no. 117517, Sep. 2023.
- [18] M. K. Mehdi, “The Influence of Infill Density and Speed of Printing on the Tensile Properties of the Three-Dimension Printing Polylactic Acid Parts,” *J. Eng. Sustain. Dev.*, vol. 27, no. 1, Jan. 2023.
- [19] M. T. Mansour, K. Tsongas, and D. Tzetzis, “Carbon-Fiber- and Nanodiamond-Reinforced PLA Hierarchical 3D-Printed Core Sandwich Structures,” *J. Compos. Sci.*, vol. 7, no. 7, Art. no. 285, Jul. 2023.
- [20] C. Zhan, M. Li, R. McCoy, L. Zhao, and W. Lu, “3D printed hierarchical re-entrant honeycombs: Enhanced mechanical properties and the underlying deformation mechanisms,” *Compos. Struct.*, vol. 290, Art. no. 115550, Jun. 2022.
- [21] A. A. Almetwally, “Multi-objective optimization of woven fabric parameters using Taguchi–Grey Relational Analysis,” *J. Nat. Fibers*, vol. 17, no. 10, pp. 1468–1478, Oct. 2020.
- [22] K. Jagatheesan and K. Babu, “Taguchi optimization of minimum quantity lubrication turning of AISI-4320 steel using biochar nanofluid,” *Biomass Conv. Bioref.*, vol. 13, pp. 927–934, Oct. 2020.
- [23] M. M. Abdulredha, S. A. Hussain, and L. C. Abdullah, “Optimization of the demulsification of water in oil emulsion via non-ionic surfactant by the response surface methods,” *J. Petrol. Sci. Eng.*, vol. 184, Art. no. 106463, Sep. 2019.



Dechawat Wannarong was born in Chonburi, Thailand in 1998. He received the Bachelor Degree in mechanical engineering from the University of Nottingham, Nottingham, United Kingdom in 2020. He also obtained another Bachelor Degree in mechanical engineering from Thammasat University, Bangkok, Thailand in 2020. In 2022, he received the Master Degree in mechanical engineering from Chulalongkorn University, Bangkok, Thailand. At present, he is pursuing a doctoral degree in the Department of Mechanical Engineering at Chulalongkorn University, Bangkok, Thailand.

His research interests include bending behaviour, the solid mechanics and the composite structures.



Thanyarat Singhanart was born in Bangkok, Thailand in 1976. She received the B.Eng in mechanical engineering from Chulalongkorn University, Bangkok, Thailand in 1998 and M.Eng in aeronautics and astronautics from Tokyo University, Tokyo, Japan in 2002 and the Ph.D. degree in aeronautics and astronautics from Tokyo University, Tokyo, Japan in 2005.

Since 1998, she has been an instructor with Department of Mechanical Engineering, Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand. Her research interests include stress analysis, fatigue, vibration, damping of composite materials and solid mechanics.